

MTH166

Lecture-3

Equations Reducible to Exact Form-I

Topic:

Equations Reducible to Exact Form

Learning Outcomes:

1. What is a Non-Exact Differential Equations.
2. What is an Integrating Factor (I.F.).
3. How to find an I.F. using different methods (Two methods).
4. How to convert a Non-exact equation into an Exact equation using I.F. and find its solution.

Non-Exact Differential Equation:

Let us consider:

$$x dy - y dx + a(x^2 + y^2) dx = 0 \quad (1)$$

Equation (1) can be re-written as:

$$(ax^2 + ay^2 - y) dx + x dy = 0$$

Comparing it with: $M dx + N dy = 0$

$$\text{Here } M = (ax^2 + ay^2 - y) \Rightarrow \frac{\partial M}{\partial y} = 2ay - 1$$

$$\text{And } N = x \Rightarrow \frac{\partial N}{\partial x} = 1$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, So, Equation (1) is Non-Exact differential equation.

Integrating Factor (I.F.):

An Integrating Factor (I.F.) is a factor which when multiplied with a non-exact differential equation, makes it an exact differential equation.

Means: $(\text{Non} - \text{Exact Equ.}) \times I.F. = \text{Exact Equ.}$

******There can be more than one I.F. for one non-exact diff. equ.

Methods to find Integrating Factor (I.F.):

1. Inspection Method:

In this method we look for a particular expression: $(xdy - ydx)$ which is non-exact.

There are many I.F. available to make this expression an exact one, like;

$$\frac{1}{x^2}, \frac{1}{y^2}, \frac{1}{xy}, \frac{1}{x^2+y^2}, \frac{1}{x^2-y^2}.$$

We know: *(Non – Exact Equ.) $\times$ I.F. = Exact Equ.*

$$1. (xdy - ydx) \times \frac{1}{x^2} = \frac{(xdy - ydx)}{x^2} = d\left(\frac{y}{x}\right) \text{ (Quotient rule)}$$

$$2. (xdy - ydx) \times \frac{1}{y^2} = \frac{-(ydx - xdy)}{y^2} = -d\left(\frac{x}{y}\right) \text{ (Quotient rule)}$$

$$3. (xdy - ydx) \times \frac{1}{xy} = \frac{(xdy - ydx)}{xy} = d\left[\log\left(\frac{y}{x}\right)\right]$$

$$4. (xdy - ydx) \times \frac{1}{x^2+y^2} = \frac{(xdy - ydx)}{x^2+y^2} = d\left[\tan^{-1}\left(\frac{y}{x}\right)\right]$$

$$5. (xdy - ydx) \times \frac{1}{x^2-y^2} = \frac{(xdy - ydx)}{x^2-y^2} = d\left[\frac{1}{2}\log\left(\frac{x+y}{x-y}\right)\right]$$

Solve the following differential equations:

Problem 1: $xdy - ydx + a(x^2 + y^2)dx = 0$

Solution: $(xdy - ydx) + a(x^2 + y^2)dx = 0$ (1)

Multiplying both sides by I.F. $= \frac{1}{x^2 + y^2}$

$$[(xdy - ydx) + a(x^2 + y^2)dx] \times \frac{1}{x^2 + y^2} = 0$$

$$\Rightarrow \frac{(xdy - ydx)}{x^2 + y^2} + adx = 0$$

$$\Rightarrow d \left[\tan^{-1} \left(\frac{y}{x} \right) \right] + adx = 0$$

Integrating both sides:

$$\int d \left[\tan^{-1} \left(\frac{y}{x} \right) \right] + a \int dx = c$$

$$\Rightarrow \tan^{-1} \left(\frac{y}{x} \right) + ax = c \text{ **Answer.**}$$

Solve the following differential equations:

Problem 2: $ydx - xdy + 3x^2y^2e^{x^3}dx = 0$

Solution: $(ydx - xdy) + 3x^2y^2e^{x^3}dx = 0$ (1)

Multiplying both sides by I.F. $= \frac{1}{y^2}$

$$[(ydx - xdy) + 3x^2y^2e^{x^3}dx] \times \frac{1}{y^2} = 0$$

$$\Rightarrow \frac{(ydx - xdy)}{y^2} + 3x^2e^{x^3}dx = 0$$

$$\Rightarrow d\left(\frac{x}{y}\right) + 3x^2e^{x^3}dx = 0$$

Integrating both sides:

$$\int d\left(\frac{x}{y}\right) + \int e^{x^3} \cdot 3x^2dx = c$$

$$[\because \int e^{f(x)} \cdot f'(x)dx = e^{f(x)} + c]$$

$$\Rightarrow \left(\frac{y}{x}\right) + e^{x^3} = c \text{ Answer.}$$

Polling Quiz

Which is the solution of $xdy + ydx = 0$:

(A) $x + y = c$

(B) $x - y = c$

(C) $xy = c$ (Answer)

(D) $x/y = c$

Methods to find Integrating Factor (I.F.):

2. If $Mdx + Ndy = 0$ is a homogeneous differential equation in x and y, then:

$$\text{I.F.} = \frac{1}{Mx+Ny}$$

Solve the following differential equations:

Problem 1. $(x^3 + y^3)dx - xy^2dy = 0$

Solution: $(x^3 + y^3)dx - xy^2dy = 0$ (1)

Comparing it with: $Mdx + Ndy = 0$

Here $M = (x^3 + y^3) \Rightarrow \frac{\partial M}{\partial y} = 3y^2$

And $N = -xy^2 \Rightarrow \frac{\partial N}{\partial x} = -y^2$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, So, Equation (1) is Non-exact .

We need to make it exact using I.F.

Look at equation: $(x^3 + y^3)dx - xy^2dy = 0$ (1)

Which is homogeneous in x and y

★ An equation is said to be homogeneous if each term of M and N is of same degree.

Here $M = (x^3 + y^3)$. Each of these two terms x and y have same degree 3.

And $N = -x^1y^2$ is also of same degree 3 as sum of powers of x and y is 3

Thus, since equation (1) is homogeneous of degree 3.

$$\text{So, I.F.} = \frac{1}{Mx + Ny}$$

$$\Rightarrow \text{I.F.} = \frac{1}{(x^3 + y^3)x + (-xy^2)y} = \frac{1}{x^4}$$

Multiplying equation (1) by I.F. to convert it to an exact equation:

$$[(x^3 + y^3)dx - xy^2dy] \times \frac{1}{x^4} = 0$$

$$\Rightarrow \left(\frac{1}{x} + \frac{y^3}{x^4}\right) dx - \frac{y^2}{x^3} dy = 0 \quad (2)$$

Which is an exact differential equation.

Solution: $\int_{y=con.} Mdx + \int (Terms\ of\ N\ free\ from\ x)dy = C$

$$\Rightarrow \int_{y=con.} \left(\frac{1}{x} + \frac{y^3}{x^4}\right) dx + 0 = c$$

$$\Rightarrow \log x - \frac{y^3}{3x^3} = c \text{ **Answer.**}$$

Problem 2. $(x^2y - 2xy^2)dx + (3x^2y - x^3)dy = 0$

Solution: $(x^2y - 2xy^2)dx + (3x^2y - x^3)dy = 0$ (1)

Comparing it with: $Mdx + Ndy = 0$

Here $M = (x^2y - 2xy^2) \Rightarrow \frac{\partial M}{\partial y} = x^2 - 4xy$

And $N = (3x^2y - x^3) \Rightarrow \frac{\partial N}{\partial x} = 6xy - 3x^2$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, So, Equation (1) is Non-exact .

We need to make it exact using I.F.

since equation (1) is homogeneous of degree 3.

So, I.F. = $\frac{1}{Mx+Ny}$

$\Rightarrow \text{I.F.} = \frac{1}{(x^2y - 2xy^2)x + (3x^2y - x^3)y} = \frac{1}{x^2y^2}$

Multiplying equation (1) by I.F. to convert it to an exact equation:

$$[(x^2y - 2xy^2)dx + (3x^2y - x^3)dy] \times \frac{1}{x^2y^2} = 0$$

$$\Rightarrow \left(\frac{1}{y} - \frac{2}{x}\right) dx + \left(\frac{3}{y} - \frac{x}{y^2}\right) dy = 0 \quad (2)$$

Which is an exact differential equation.

$$\text{Solution: } \int_{y=\text{con.}} M dx + \int (\text{Terms of } N \text{ free from } x) dy = C$$

$$\Rightarrow \int_{y=\text{con.}} \left(\frac{1}{y} - \frac{2}{x}\right) dx + \int \frac{3}{y} dy = c$$

$$\Rightarrow \frac{x}{y} - 2 \log x + 3 \log y = c \quad \mathbf{Answer.}$$

